

Tools for Recovering and Visualizing Intrasite Spatial Data: Interpolation and Spatial Autocorrelation.

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Intra-Site Spatial Data Recovery and Visualization

Historical Issues:

1. Locations of activities – what happened where and why?
2. Locations of social groups – who lived where and why?
 - class, gender, ethnicity, etc.

Sources of Data: Archaeological Spatial Structure

1. patterns in the location of features: ***site structure***
 - buildings
 - pits
 - fences
2. patterns in the horizontal distribution of artifacts

Patterns in the horizontal distribution of artifacts

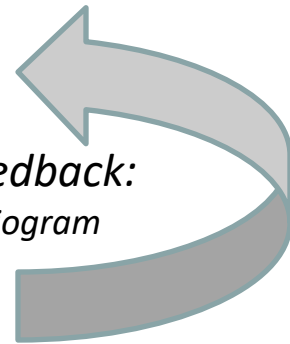
1. The spatial data recovery process

- Model how we collect spatial data.

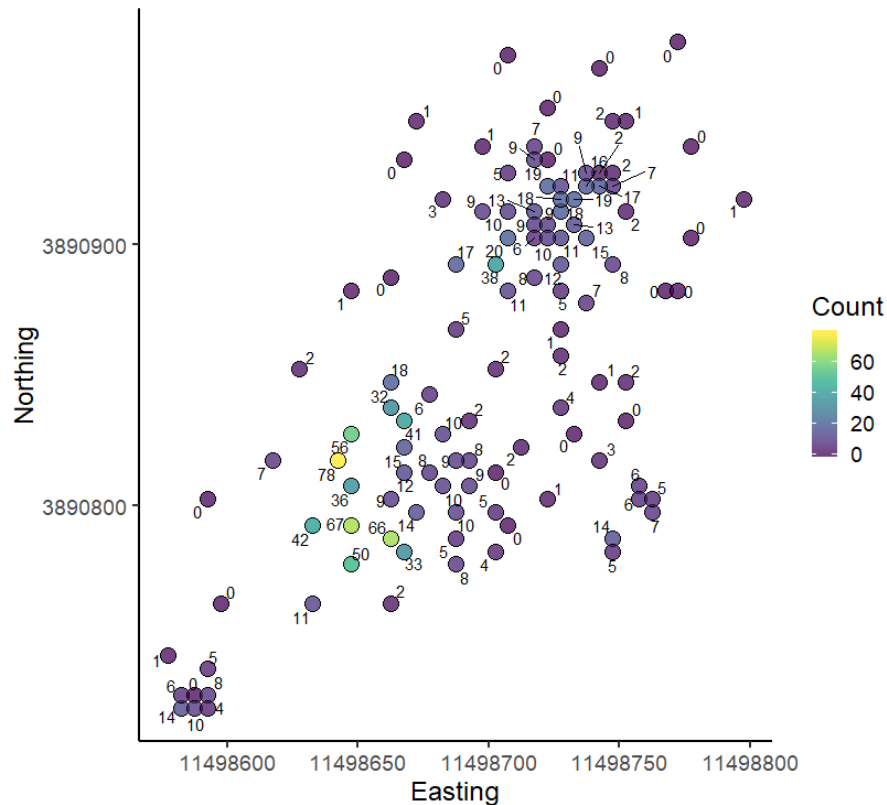
2. The analysis process

- Mapping raw data and statistical models of them.

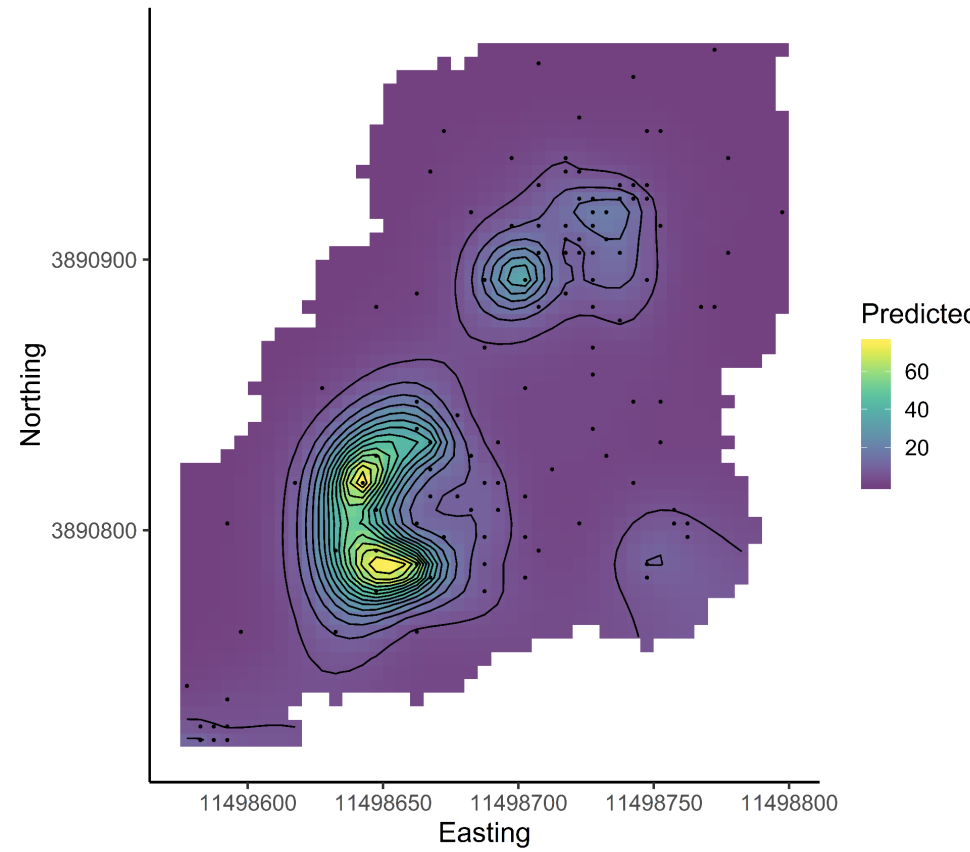
Feedback:
Variogram



Creamware



Creamware



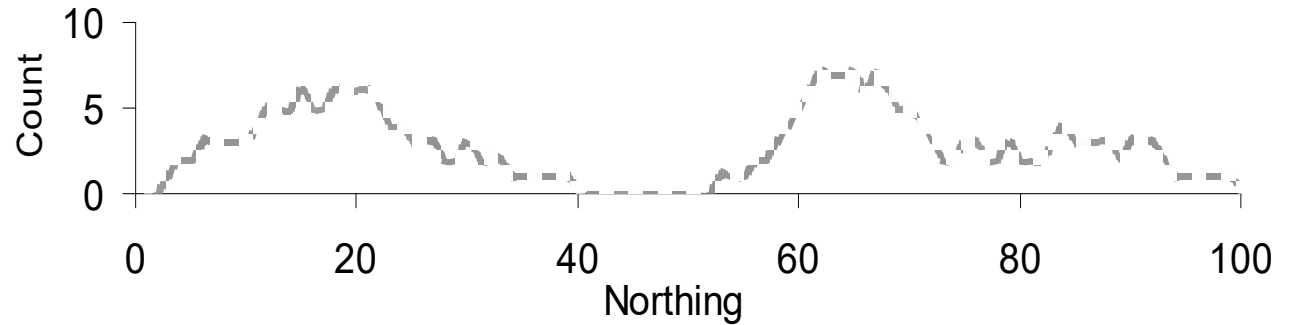
Recovery

The Point Process:



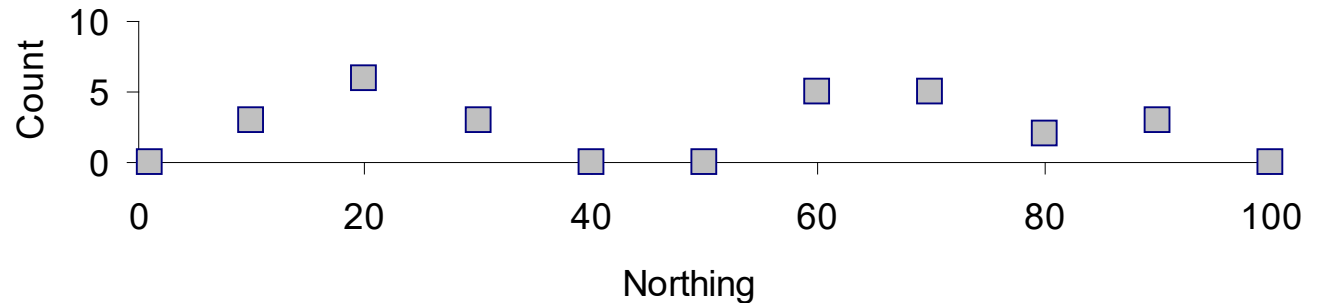
The Moving-Average
Process:

(quadrat diameter=10)



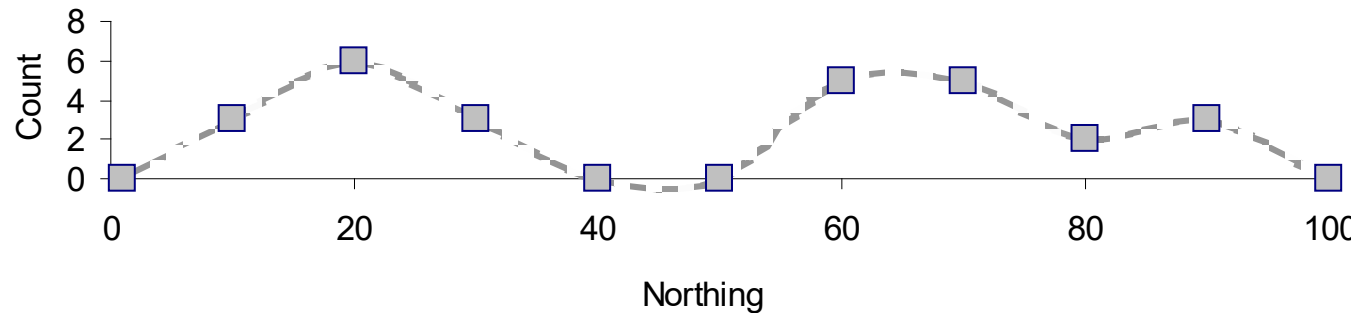
Sample the M-A
Process

(quadrat spacing =10)



Analysis

Estimate the
M-A Process
from the sample:



Important Decisions in Designing a Spatial Sampling Strategy

Quadrat size

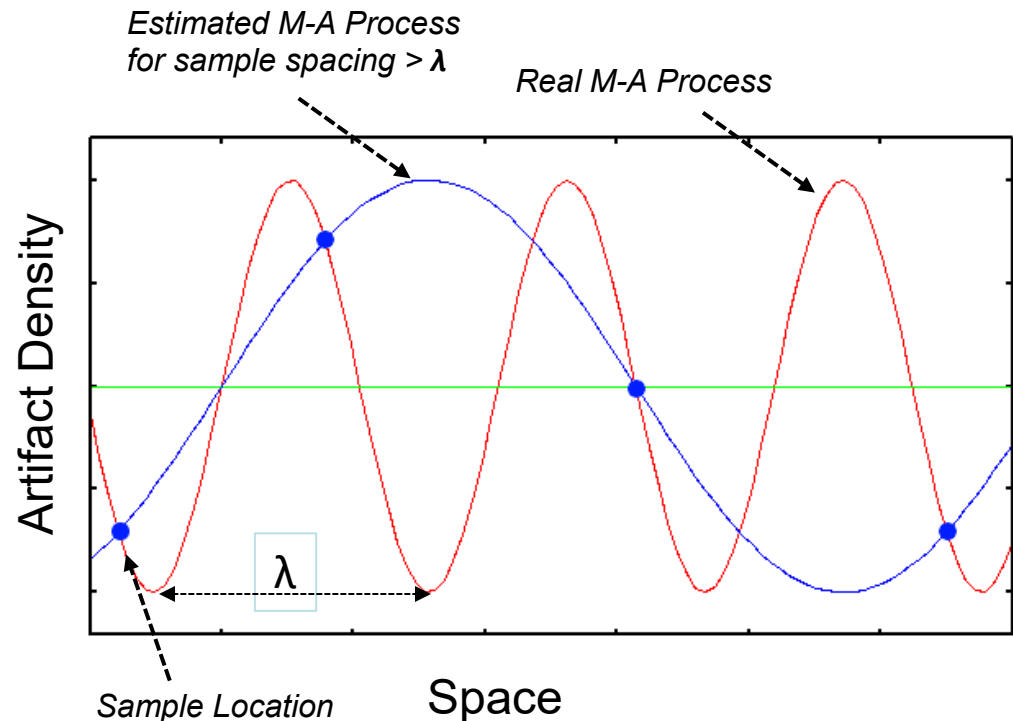
- Determines the minimum possible quadrat spacing
e.g. with 5-foot quadrats, quadrats centers can be no less than 5 feet apart

Quadrat spacing

- Determines the *minimum size* of the of the spatial phenomena we can reliably see.

Help from signal processing

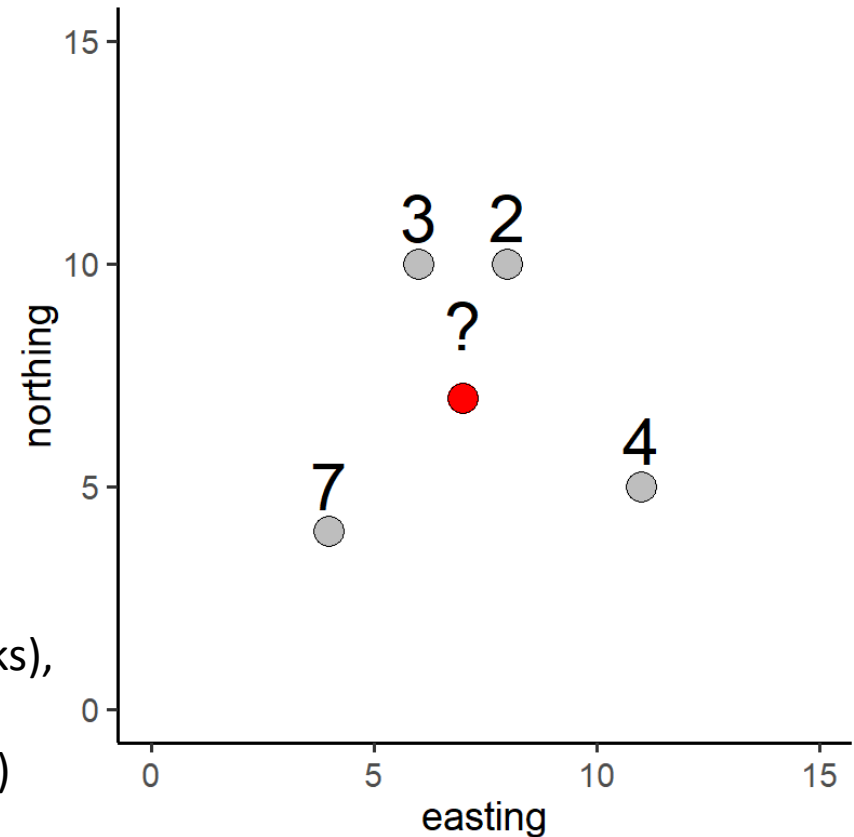
- Assume artifact densities (the M-A process) follow a sine wave in 1-D space, with wavelength λ .
- To be able to reconstruct the sine wave from samples, **sample spacing must be $< \lambda/2$** . *e.g.* 10-foot wavelength requires sample spacing less than 5 feet.
- The sine wave is worst-case scenario.



Interpolation

Many methods...

- Inverse distance weighting (IDW)
- Kriging
- Others
 - TINs (triangulated irregular networks),
 - splines
 - polynomial regression (local, global)



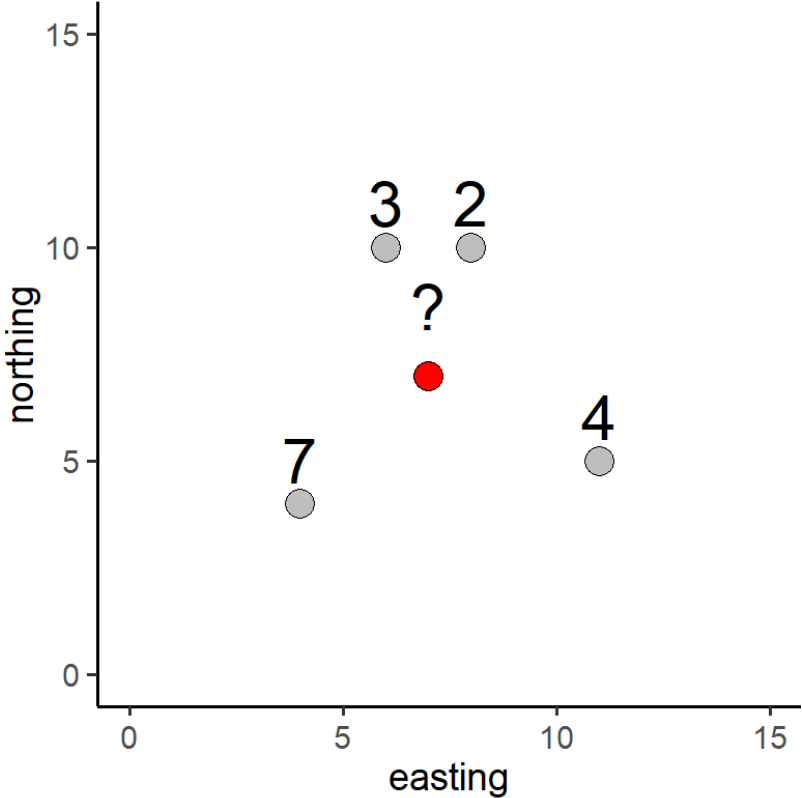
IDW and Kriging both make estimates of value of the z variable at an unsampled point in (x,y) space, as a ***weighted average of the values at nearby points, where z values are known.***

So....

$$\hat{z}_j = \frac{\sum_{i=1}^n w_i z_i}{\sum_{i=1}^n w_i}$$

Inverse Distance Weighting

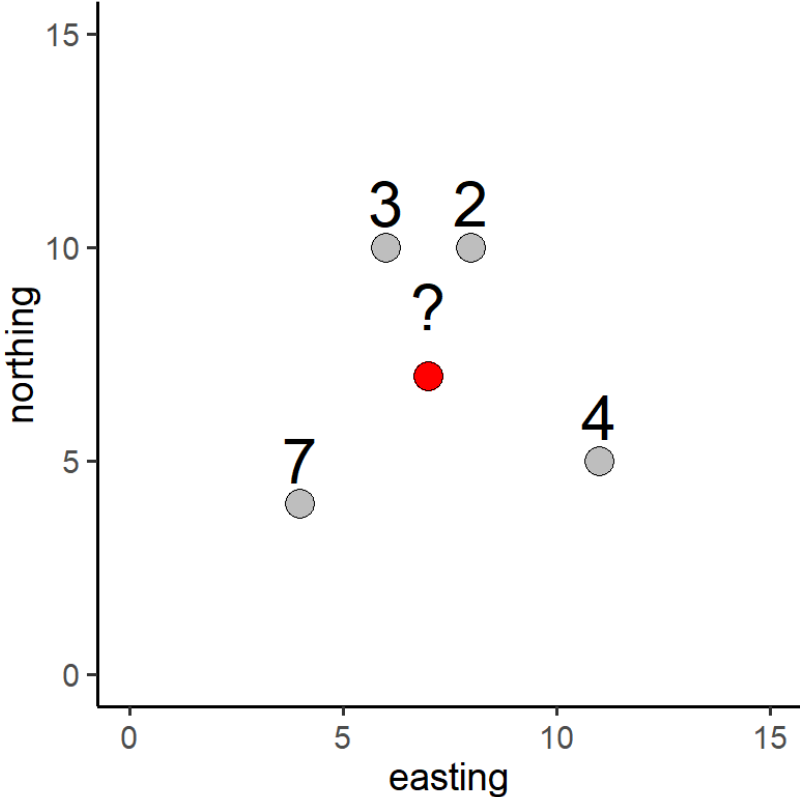
$$w_i = \frac{1}{d_{ij}^p}$$



Point ID	northing	easting	z	$Distance = d_{ij}$	$w_i = 1/d_{ij}$	$w_i * z$
1	11	5	4	4.47	0.22	0.89
2	6	10	3	3.16	0.32	0.95
3	8	10	2	3.16	0.32	0.63
4	4	4	7	4.24	0.24	1.65
Sum					1.09	4.13
5	7	7	?			

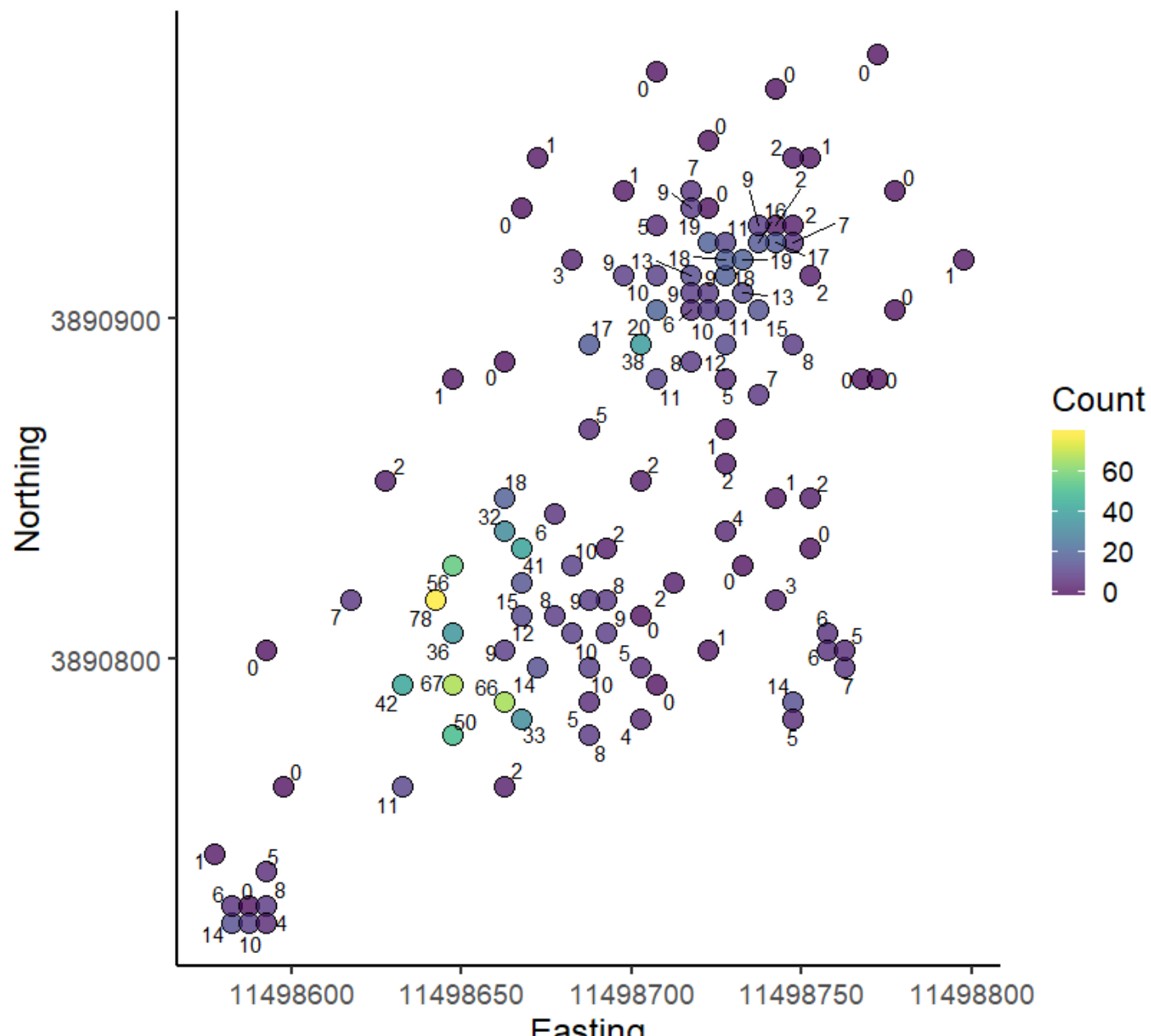
Inverse Distance Weighting

$$w_i = \frac{1}{d_{ij}^p}$$



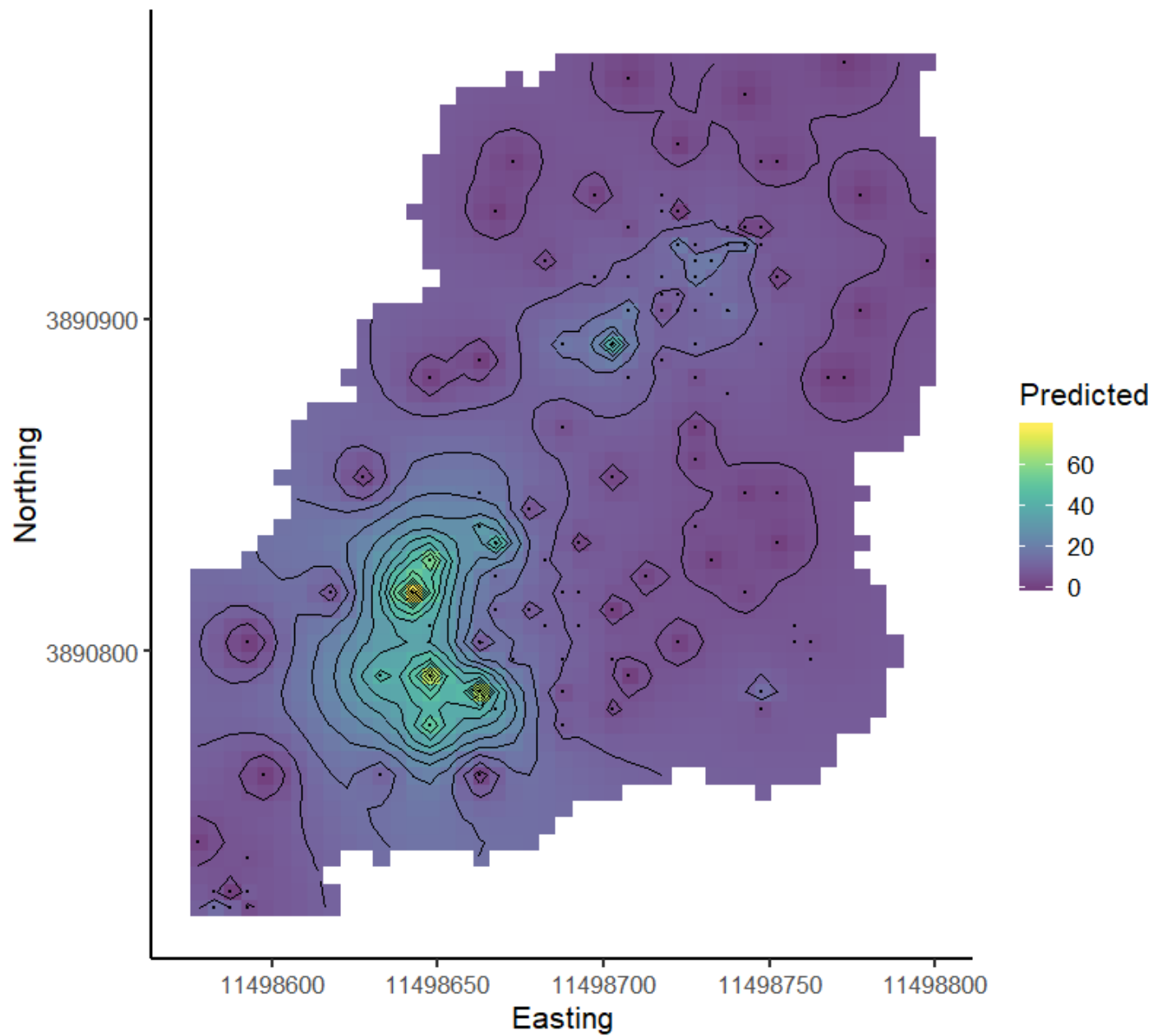
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Sum					1.09	4.13
5	7	7	$4.13/1.09 = 3.8$			

Creamware



Creamware

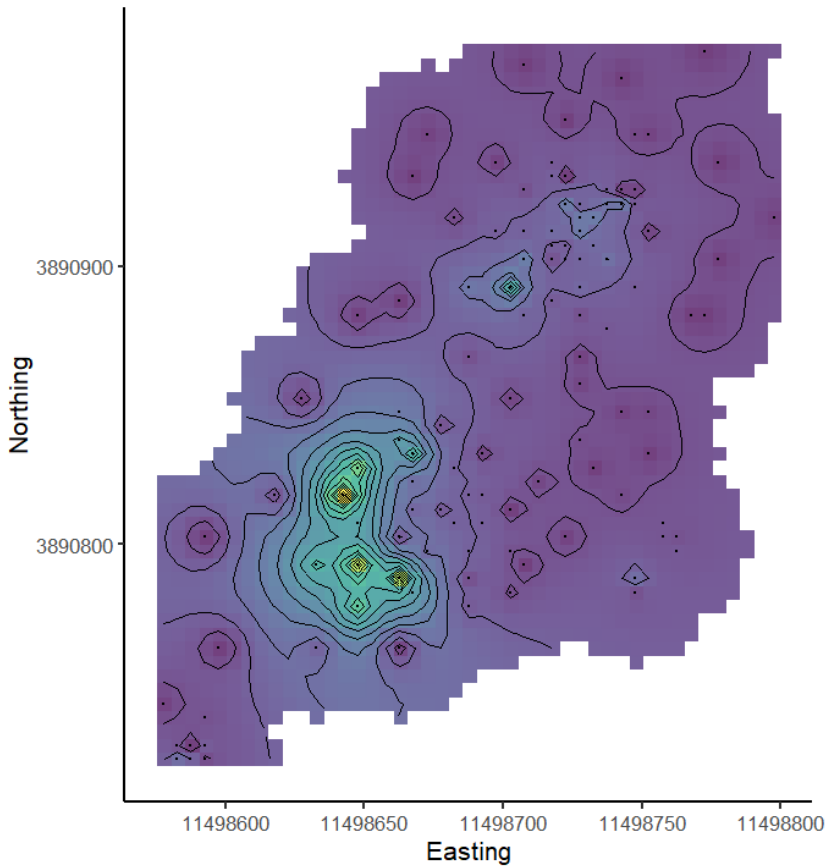
IDW, $p=2$



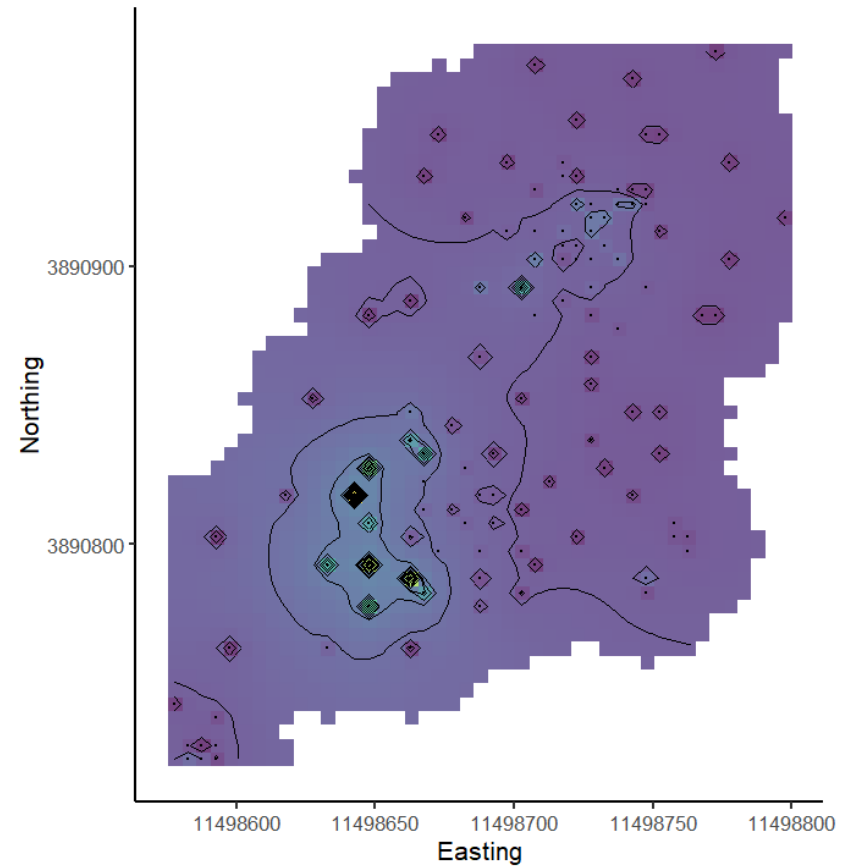
Pesky Questions about IDW

- what value for p ?

Creamware
IDW, $p=2$



Creamware
IDW, $p=1$

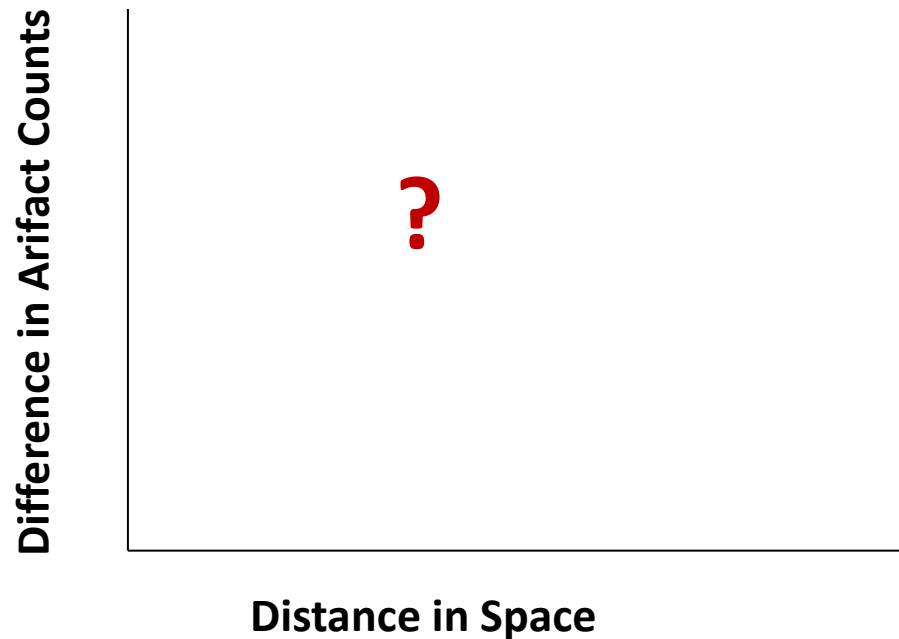


Doing Better than IDW

- ***p*** should depend on the manner in which ***differences*** between z-values increase with ***distances*** between x,y coordinates....

“Spatial autocorrelation”

To what extent do quadrats that are farther apart in 2-d space (*e.g.* Easting and Northing) tend to have variable values (*e.g. artifact counts*) that are more different.



Kriging (after D.R Krige)

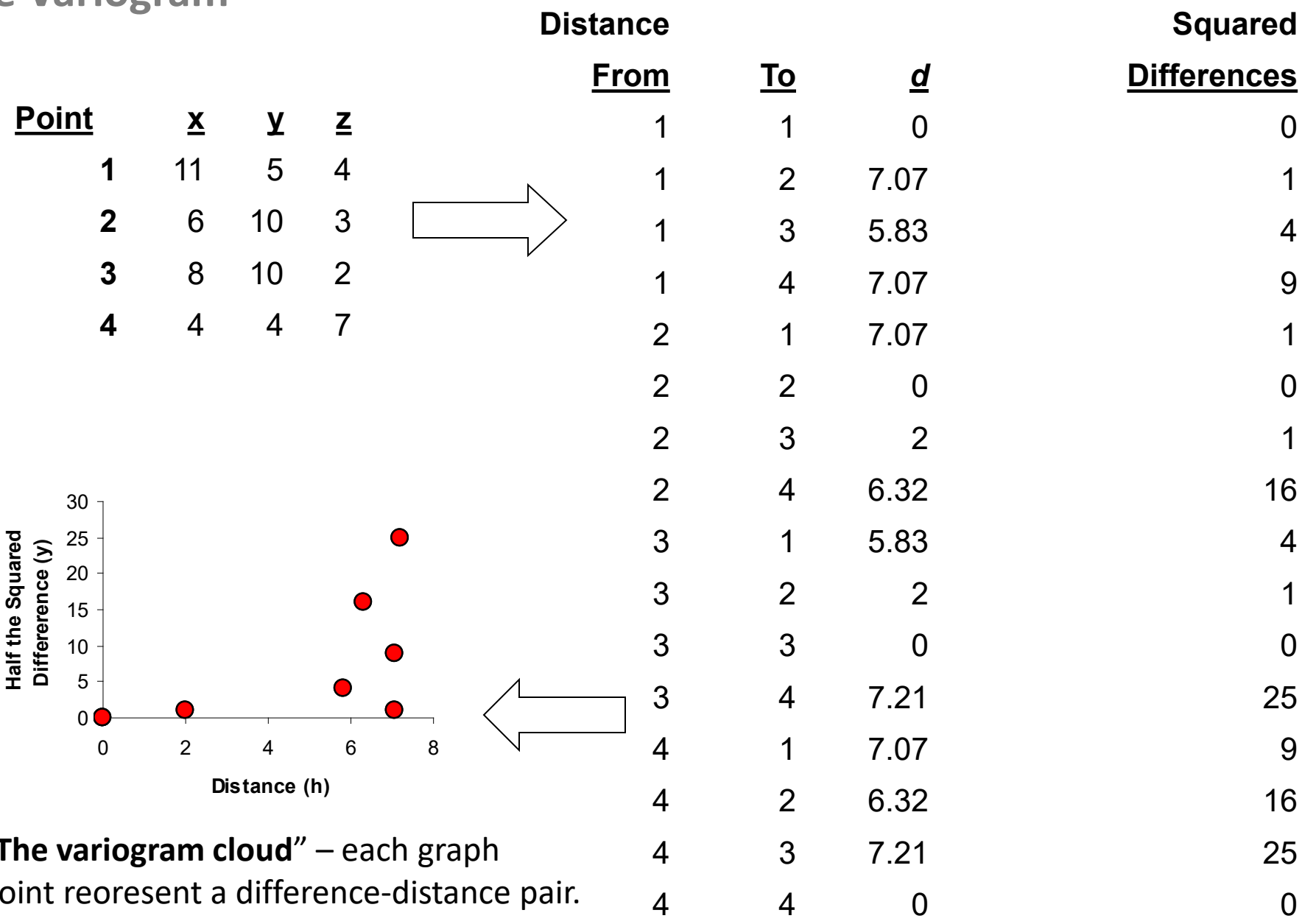
A weighted-averaging interpolation method in which the ***weights depend on the spatial autocorrelation structure of the data***, AND that produces estimates of Z that are designed to minimize mean-squared prediction error.

Variogram

The graphical tool we use to measure the autocorrelation structure of spatial data.



The Variogram

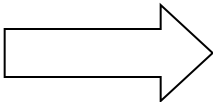


The Variogram

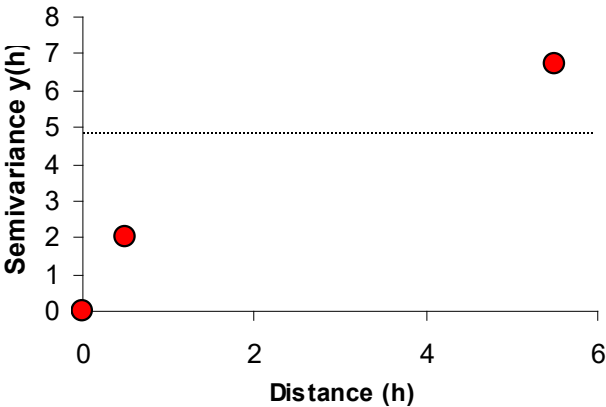
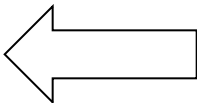
<u>Point</u>	<u>x</u>	<u>y</u>	<u>z</u>
1	11	5	4
2	6	10	3
3	8	10	2
4	4	4	7

Variance

4.67



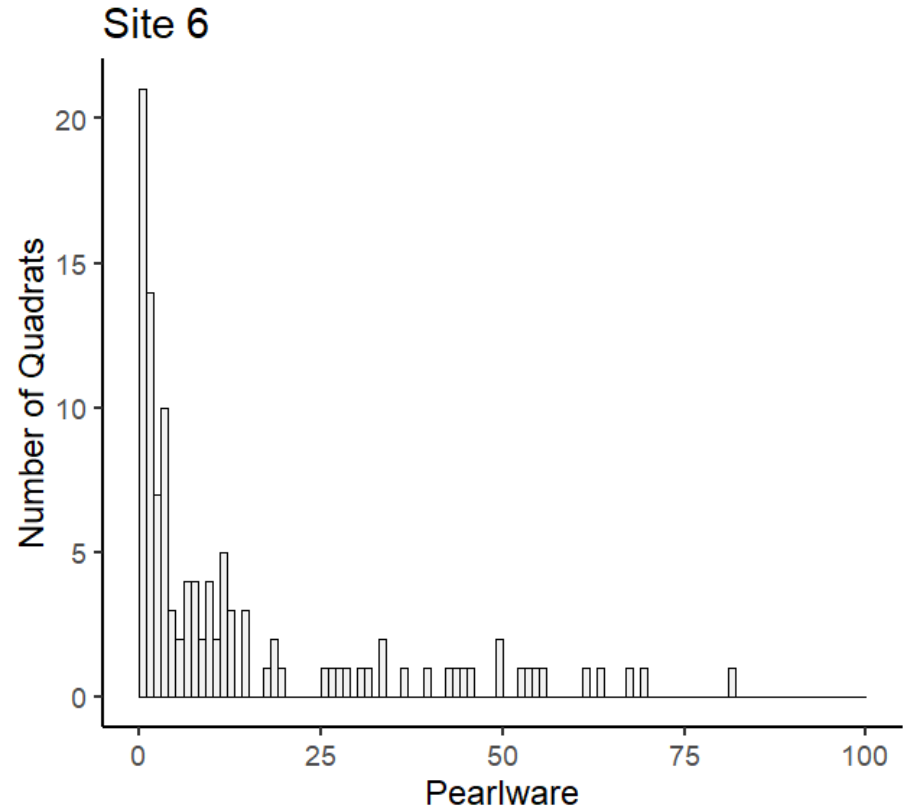
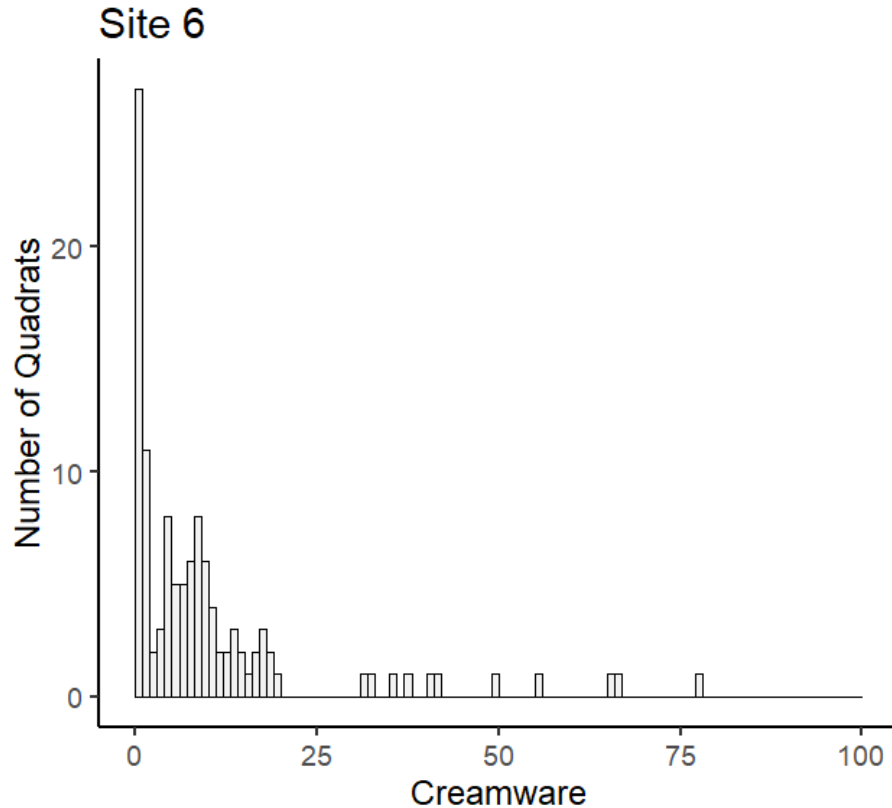
<u>Distance</u>			<u>Squared</u>	<u>Distance</u>		
<u>From</u>	<u>To</u>	<u>h</u>	<u>Differences</u>	<u>Class</u>	<u>y(h)</u>	<u>mean(h)</u>
1	1	0	0	0		
2	2	0	0	0		
3	3	0	0	0		
4	4	0	0	0	0	0
2	3	2	1	1-5		
3	2	2	1	1-5	0.5	2
1	3	5.831	4	5-10		
3	1	5.831	4	5-10		
2	4	6.325	16	5-10		
4	2	6.325	16	5-10		
1	2	7.071	1	5-10		
1	4	7.071	9	5-10		
2	1	7.071	1	5-10		
4	1	7.071	9	5-10		
3	4	7.211	25	5-10		
4	3	7.211	25	5-10	5.5	6.70175



“The variogram” – each graph point represents the means of several difference-distance pairs.

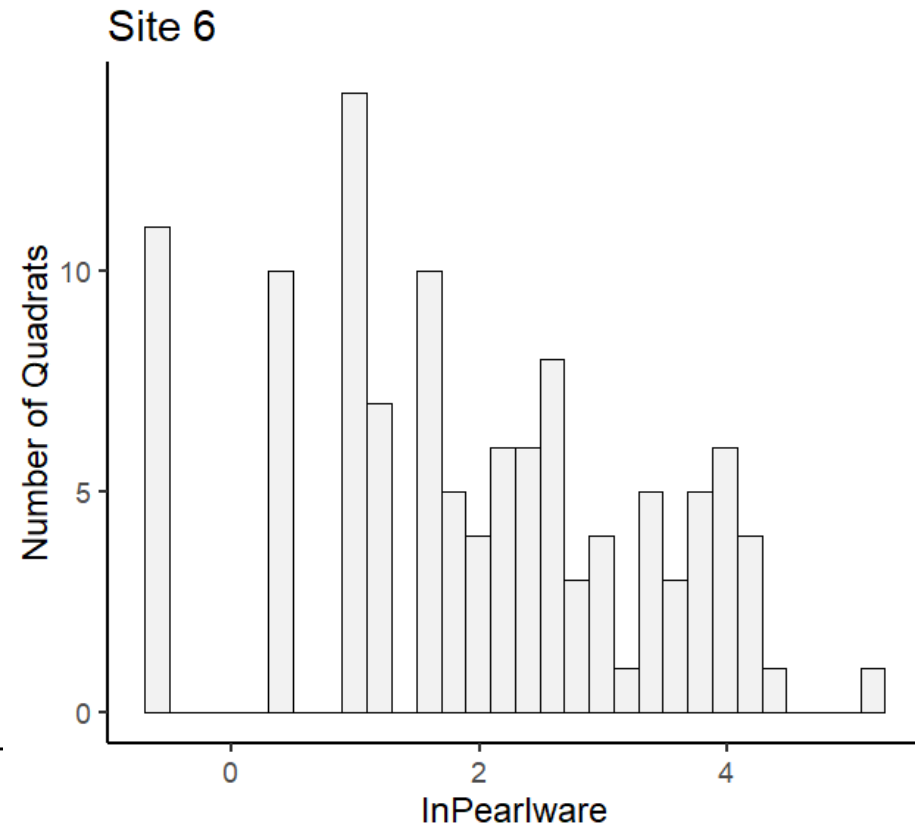
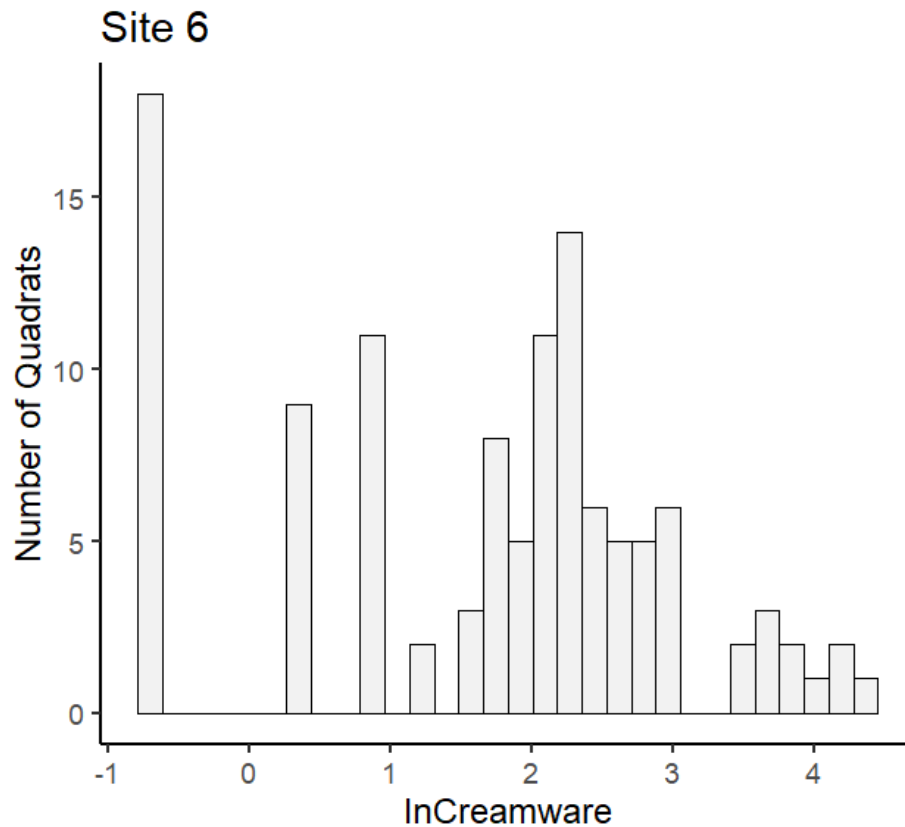
The Variogram

- The mathematical model behind the variogram and kriging assumes that the spatially distributed variable has a normal or Gaussian distribution.
- But artifact counts always have long right tails...



The Variogram

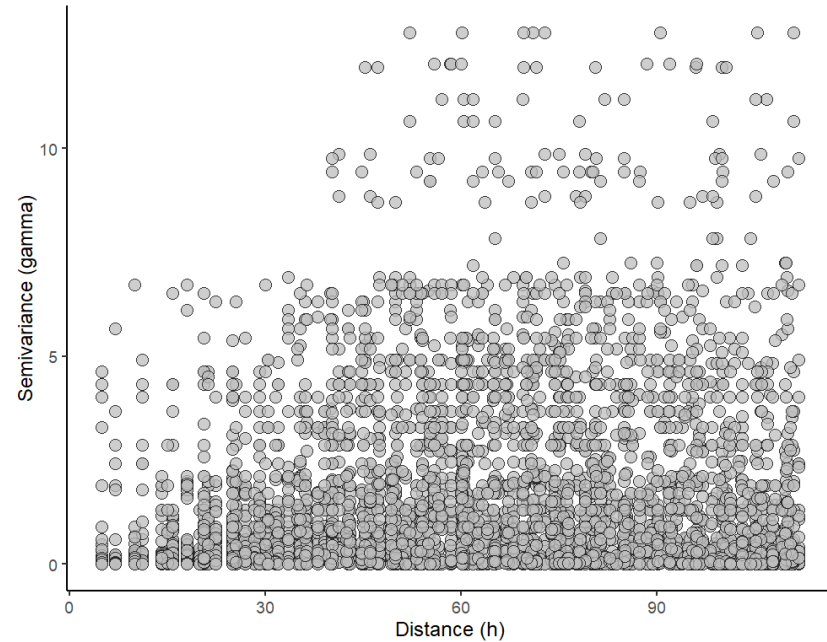
- Transforming the counts to a log scale helps.
- Because $\ln(0)$ is undefined, we take logs of “started counts”
 - e.g. $\ln(\text{Creamware} + .5)$



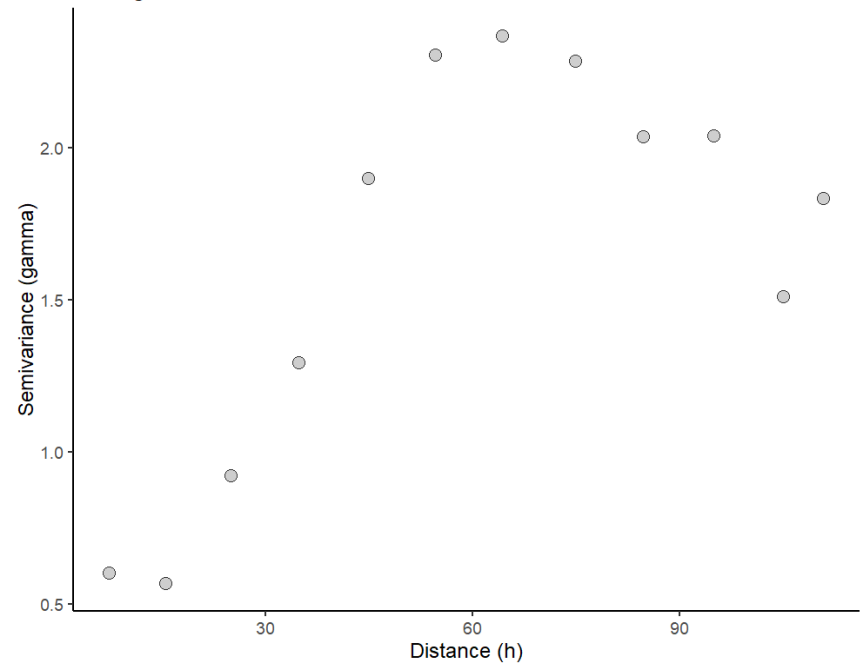
The Variogram

- Transforming the counts to a log scale helps.
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 - e.g. $\ln(\text{Creamware} + .5)$

Variogram Cloud: InCreamware



Variogram: InCreamware



The Variogram

$$\gamma_h = \frac{1}{2n(h)} \sum_{i=1}^n (z(x_i, y_i) - z((x_i, y_i) + h))^2$$

Add up all the difference in pairs of z values

Square the differences in pairs of z values

The semivariance for distance class h

$n(h)$ = the number of point pairs that fall in distance class h

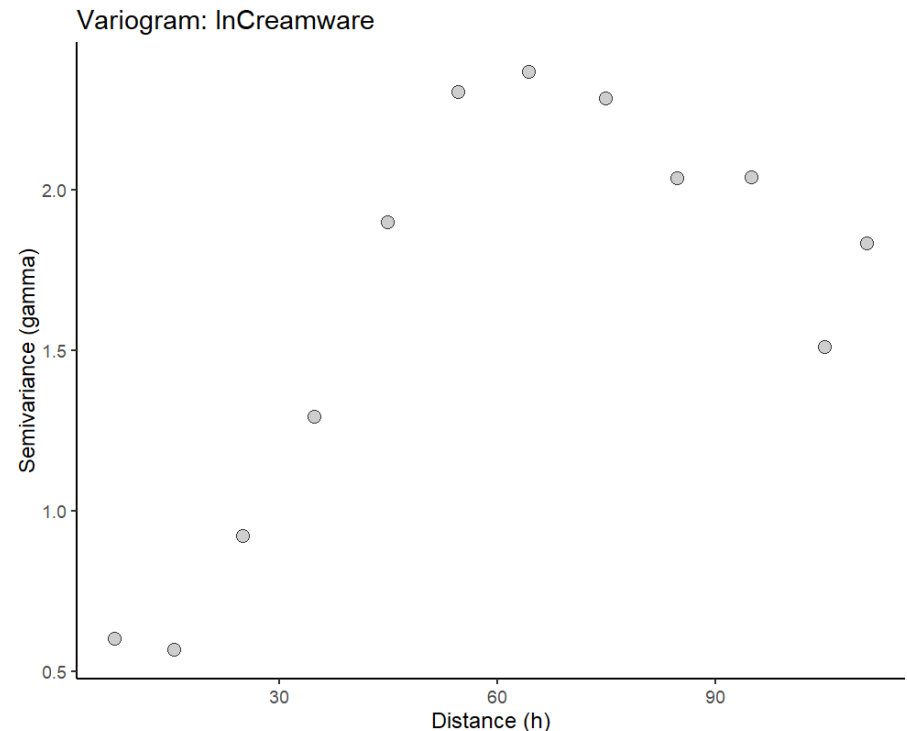
$z(x_i, y_i)$ = the value of z at the i 'th point with coordinates x, y

$z(x_i, y_i) + h$ = the value of z at the i 'th point that is h away from the point with coordinates x, y

(x, y) are 2-d spatial coordinates (easting, northing)

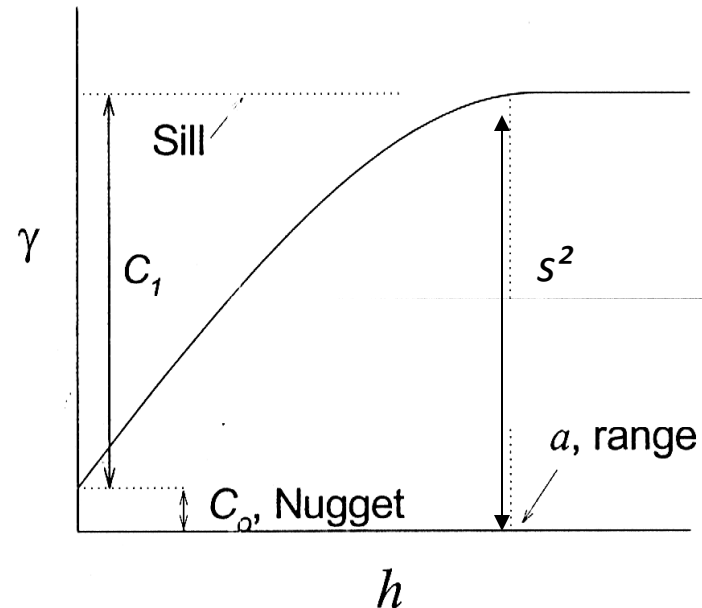
z is the variable value (artifact counts)

h is a distance and direction vector:
"all the points that are a certain distance apart from the i 'th x, y pair".



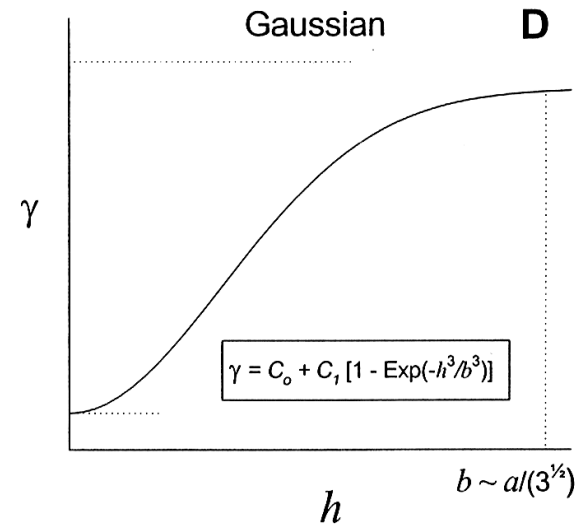
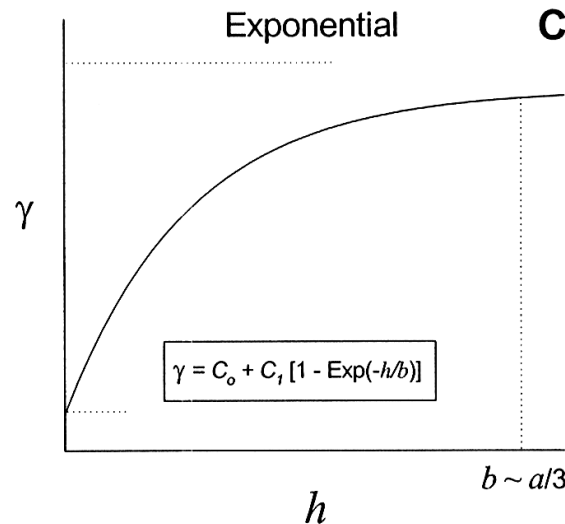
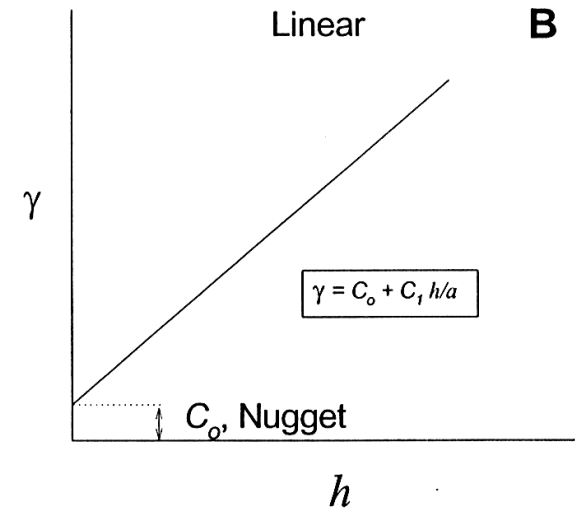
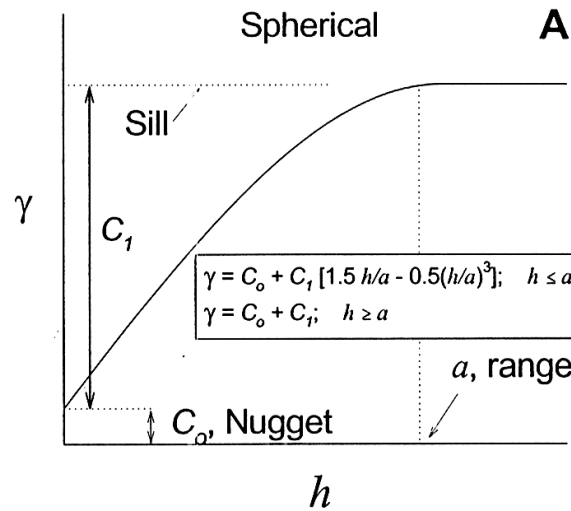
Variogram Lingo

- **Sill:** for larger values of h the variogram levels out, indicating that there no longer is any auto correlation between data points.
- *If the data are “well behaved” (Gaussian and stationary) the sill should be equal to the **variance** (s^2) of the z values.*
- **Range:** is the value of h where the sill occurs (or 95% of the value of the sill). This is the distance beyond which pairs of values are no longer autocorrelated.
- **Nugget variance:** a non-zero value for *gamma* when $h = 0$. Produced by various sources of unexplained error (e.g. measurement error).



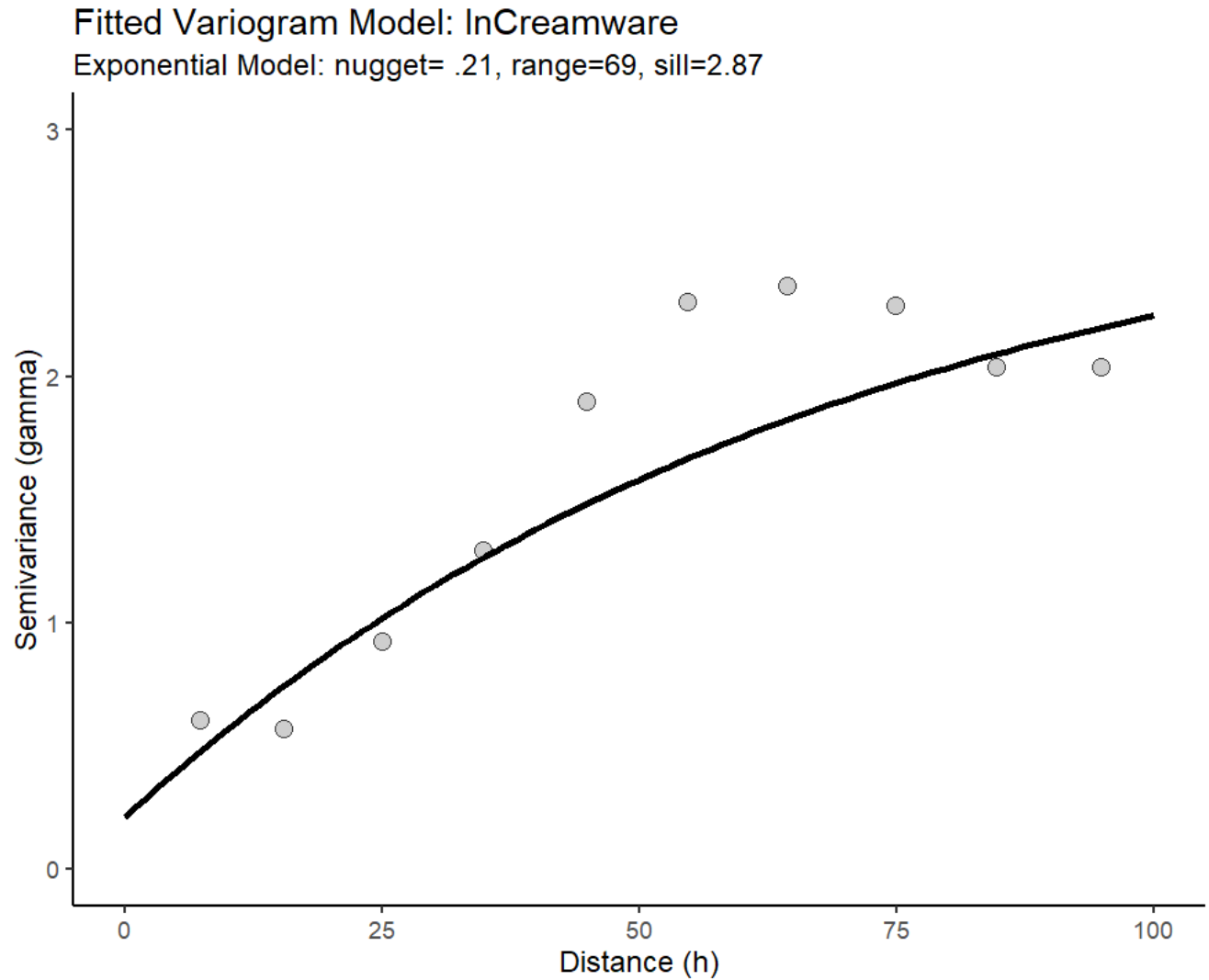
The Variogram

Variogram Models:
Differently shaped
curves, defined by
different equations.



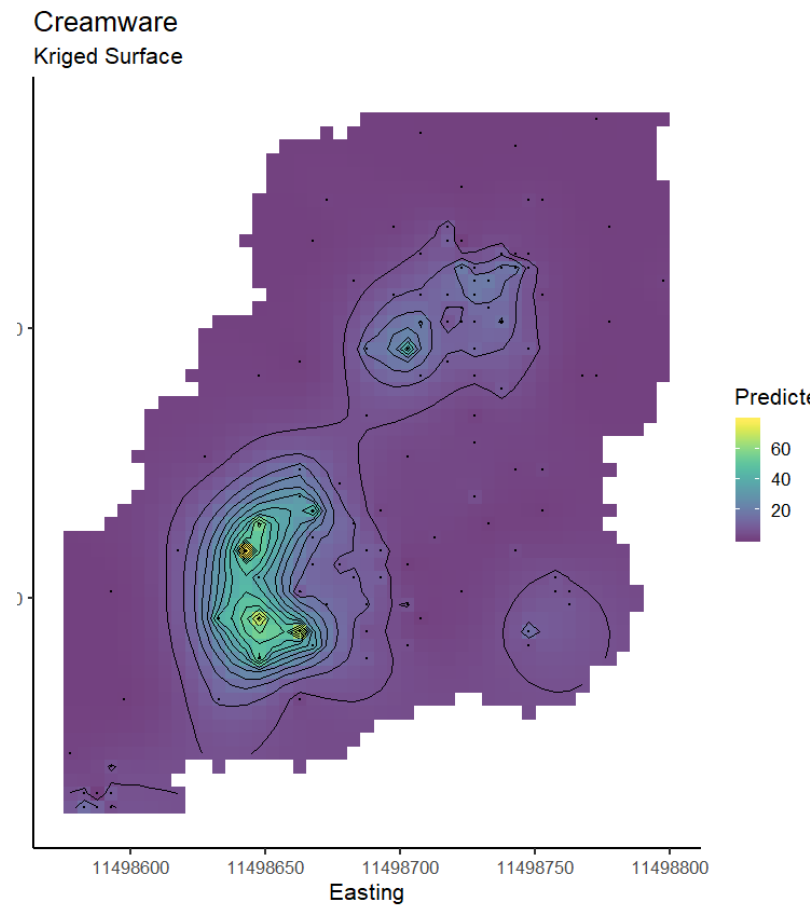
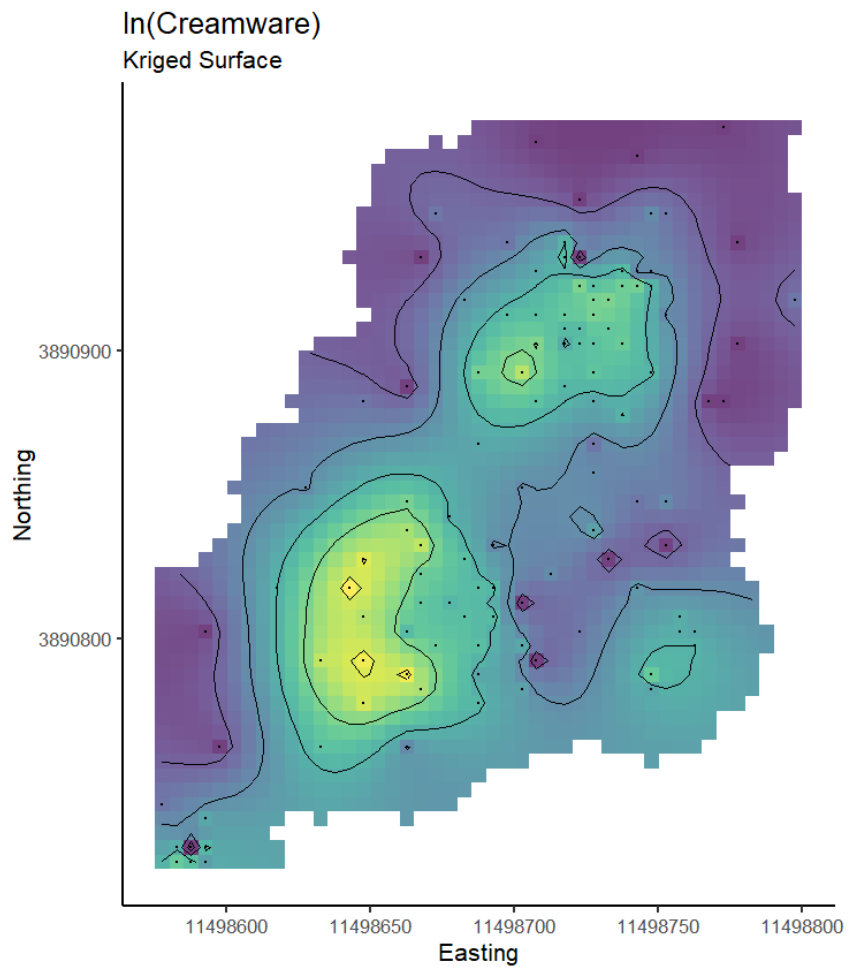
The Variogram

Variogram Models:
Differently shaped
curves, defined by
different equations.



Site 6

Kriged Density Surfaces



The variogram is a useful spatial data analysis tool !!

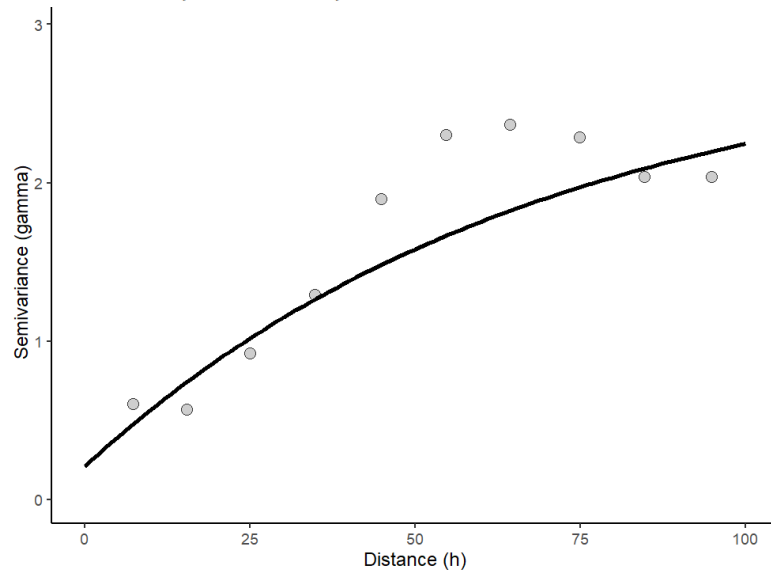
You can use it during and excavation to see if your spatial sampling strategy is sufficient to capture spatial patterning ***at some scale***.

- Quadrat size (too small?)
- Quadrat spacing (too far part?)
- ***Given quadrat size and spacing is interpolation reasonable?***

Site 6

Empirical Variogram with Fitted Model

Creamware: ln(started counts)



Site 30

Empirical Variogram with Fitted Model

Creamware: ln(started counts)

